

International Islamic University Chittagong
Department of Computer Science and Engineering
Report on Final Year Project/Thesis as Complex Engineering Problem

The following report will justify that the work done in your project/thesis is a solution to a complex engineering problem. To do this, firstly, specify which attributes of complex engineering problem (P) are present in the problem you have formulated and solved. Secondly, specify which complex engineering activities (A) are involved when solving the problem, particularly when you required to communicate for solving the problem. Provide explanation for each P and A involved. Finally, give an overall comment justifying that you have solved a complex engineering problem in your project/thesis.

Title of the Project/Thesis:	CacheFlow: Memory-Aware Expert Scheduling for Efficient Mixture-of-Experts Models	
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Which attributes of the Complex Engineering Problems (P) are present?

Attribute of Complex Engineering Problem	Explanation (Justify that the attribute was present and you have addressed it sufficiently)
P1: Depth of knowledge required	This thesis involved an extremely broad range of knowledge in deeply complex areas such as both Transformer-based LLM and Mixture-of-Experts (MoE) model architecture, Operating System-level memory management and control, and low-level GPU hardware mechanisms. Solution development required integrating an understanding of asynchronous PCIe data transfers, allocating pinned (page-locked) CPU memory, disk I/O optimization (Safetensors zero-copy slicing), and parallel computation through CUDA Streams and Events-all of which lie well beyond the scope of standard undergraduate CS curricula.
P2: Range of conflicting requirements	The research solved a critical multi-dimensional trade-off between minimal GPU VRAM usage and bounded inference latency degradation. Simply loading all experts to disk causes dramatic sequential pipeline stalls, whereas all experts in VRAM causes inevitable Out-Of-Memory (OOM) conditions. CacheFlow managed to navigate this boundary by implementing a Pareto-

	optimal curve through asynchronous disk I/O, pinned staging memory, and GPU compute.
P3: Depth of analysis required (no obvious solution)	This represented a significant non-trivial challenge; there were no "off-the-shelf" solutions for pipelined, asynchronous MoE inference on a single GPU, as typical MoE architectures are inherently synchronous (DeepSpeed MoE) or carry extreme synchronous latencies (MoE-Infinity). In-depth analysis of the execution path, injection of asynchronous data prefetchers, and construction of a custom ExpertCache with intra-batch freezing logic were required to avoid race conditions and deadlocks.
P4: Familiarity of issues (infrequently encountered issues)	The project tackled extremely complex and relatively uncommon systems-engineering challenges. Orchestration of parallel Disk->RAM->GPU data pipelines and asynchronous PCIe data transfer mechanisms were key design problems, as were dealing with the race conditions necessary to prevent the cache evicting an expert being processed in the forward pass by an in-batch query-tasks which are difficult to encounter outside of extremely advanced industrial R&D environments.
P5: Extent of applicable codes	Following rigorous software engineering and machine learning methodology was essential for building the system. Object-oriented design with clear module separation was used, and adherence to the PyTorch API was enforced throughout. Safe CUDA memory management techniques to avoid memory leaks, and strict adherence to standard ML evaluation practices (e.g. Floating-point tolerances) were enforced.
P6: Extent of stakeholder involvement and conflicting requirements	The work successfully satisfied the needs of a wide variety of stakeholders: the AI community requires running the newest large 60+ expert models, while hardware developers must deploy on affordably priced, available consumer-grade hardware, and institutions must find the lowest computing costs. CacheFlow meets all three by deploying state-of-the-art, enormous models on commonplace hardware (e.g. RTX 3060 12GB).
P7: Interdependence	Each part of the architectural system was heavily reliant on others. Variables such as the allocation of the staging buffer S , the GPU cache size C , the data transfer throughput on PCIe, and the entropy of the routing information were all interconnected. By changing C , a non-linear change to system throughput, latency, and maximal VRAM usage was produced by raising the GPU cache hit rate and thus the number of asynchronous disk fetches.

Which attributes of Complex Engineering Activities (A) are involved?

Attribute of Complex Engineering Activities	Explanation (Justify that the attribute was present and you have addressed it sufficiently)
A1: Range of resources	The research synthesized several niche resources: many distinct kinds of NVIDIA GPU (T4, H100), the low-level PyTorch CUDA APIs (torch.cuda.Streams, torch.cuda.Events), a fast disk slicing library (Safetensors), the Hugging Face transformers library for the core architectural components of MoE LLMs, and state-of-the-art literature pertaining to MoE LLM memory optimization techniques.
A2: Level of interactions	Constant team coordination and intensive discussion with the supervisor were crucial for architecting the 3-tier design, debugging complex asynchronous race conditions and for evaluating detailed, multi-dimensional performance metrics. The extensive use of high-level technical communication was instrumental in confirming numerical accuracy of the model transformations and analyzing the multi-dimensional plots produced.
A3: Innovation	This project's innovation lies in its unique three-tier, Disk -> Pinned RAM -> GPU VRAM memory architecture with asynchronous data prefetching and an LRU intra-batch freezing policy. Modifying off-the-shelf expert modules and the associated infrastructure to natively implement asynchronous I/O, while fundamentally maintaining the original underlying model structure is a very novel and elegant engineering feat.
A4: Consequences to society and the environment	The research democratized the deployment of massive MoE models onto affordable, common consumer hardware-away from energy-intensive, expensive cloud computing clusters. In doing so, it promotes greener and more sustainable AI research and development while granting access to cutting-edge LLMs for researchers lacking institutional resources.
A5: Familiarity	None of the system engineering actions taken had been performed by the team before. The core novelty and difficulty lay in engineering a custom asynchronous memory controller that intercepting data transfers with low-level CUDA events, building safeguards into the cache eviction policy to prevent concurrent data access during execution on the GPU, all of which involved significant self-directed study of system fundamentals.

Comments:

This thesis fully resolves an extremely challenging engineering problem involving complex architectural design, conflicting constraints and a system-wide innovation. The work on CacheFlow confirms that a Mixture-of-Experts models' memory bottleneck may be successfully alleviated by an asynchronous three-level memory hierarchy. While only a 5-9 fold decrease in peak GPU VRAM is obtained (e.g. 988 MB when C=1), it does so without

compromising the correctness of computations through the intra-batch freezing, thereby satisfying all requirements of the Complex Engineering Problem and thus representing an outstanding, highly precise scientific achievement.

Approval of Supervisor	Approval of Examination Committee
Signature	Signature